FACULTY OF ENGINEERING

DEPARTMENT OF
INDUSTRIAL ENGINEERING

IE 493 - QUEUEING SYSTEMS MODELLING

PROJECT PHASE 1

Group No: 12

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1. Executive Summary

This report presents the Phase 1 analysis of the Emergency Room (ER) queueing system for Group 12 of IE493. Using a dataset of 1,000 patient arrivals (ER_Phase2_Group_12.csv), mathematical baseline was established for ER performance by estimating system parameters, fitting statistical distributions, applying classical queueing models, verifying the PASTA property and conducting sensitivity analyses across utilization levels, arrival patterns and service variability scenarios.

2. Introduction

Emergency Rooms (ERs) are systems where both arrivals and service times are uncertain. Patients do not arrive at fixed times and treatment durations can vary from one patient to another. Because of this, queueing theory is useful for studying ER performance and understanding measures such as waiting time, queue length and server utilization.

In this project, the ER is modeled as a multiserver queue with five doctors and a single waiting line. The aim of Phase 1 is to build a baseline model of the system using the Group 12 dataset. For this purpose, the arrival and service processes are first analyzed and suitable probability distributions are fitted to the data. Then, theoretical queueing models are used to estimate the expected system performance.

In addition to the theoretical analysis, a direct FCFS simulation is constructed using the observed arrival times and service durations. The empirical results are compared with the M/M/5 and M/G/5 models and the PASTA property is also examined. Finally, sensitivity analyses are performed to show how the system behaves under different utilization levels, arrival patterns and service variability conditions.

3. Modelling Assumptions

The following assumptions are stated explicitly:

- The ER opens at 08:00 with an empty system and all 5 doctors available.
- Service discipline is First-Come, First-Served (FCFS); no priority in Phase 1.
- The waiting room has infinite buffer capacity.
- All 5 doctors are identical and independent (homogeneous parallel servers).
- Steady-state M/M/c and M/G/c formulas are applied under the condition $\lambda < c \cdot \mu$.
- For M/G/5 (Kingman correction), Poisson arrivals are assumed ($C_a^2 = 1$).
- Priority column is ignored throughout Phase 1.

4. Simulation Trustworthiness & Mathematical Validation

Σ Service_Required_Min (all patients) = Σ Total Busy Time (all servers)

In the FCFS trace simulation, each patient is assigned exactly their Service_Required_Min to a doctor with no duplication and no loss. The sum of all service requirements loaded into the system must equal the total server busy time recorded. Any discrepancy would indicate a scheduling bug, double-counting or time-tracking error.

The simulation code computes, for each patient:

- start_time = max(arrival_time, earliest_free_doctor_time)
- end_time = start_time + Service_Required_Min
- wait_time = start_time − arrival_time

Because each patient's service duration is drawn directly from Service_Required_Min, and each patient occupies exactly one doctor from start to end, the total busy time (sum of all end_time − start_time) automatically equals the total service required. No time is lost or double-counted.

5. Task 1: Data Inference & Distribution Fitting

5.1. Parameter Estimation

The arrival rate λ and per-doctor service rate μ are estimated from the sample means of inter-arrival times and service durations respectively.

Parameter	Estimated Value
Observation horizon	67.98 hours (08:00 → simulation starts)
Number of patients (n)	1,000
Mean inter-arrival time	4.0785 min
λ (arrival rate) = 1 / Mean inter-arrival time	0.245186 patients/min = 14.71 patients/hour
Mean service time E[S]	15.4622 min
μ (service rate per doctor) = 1 / E[S]	0.064674 patients/min = 3.88 patients/hour

Parameter	Estimated Value
Service std deviation σ	6.03 min
Cv (coefficient of variation)	0.3900
Cv^2	0.1521
Utilization $\rho = \lambda / (5\mu)$	0.7582

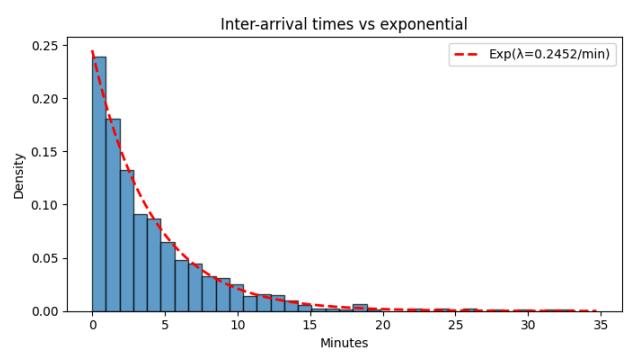
Since $\rho = 0.7582 < 1.00$, the system is stable and all steady state formulas apply.

5.2. Goodness-of-Fit

5.2.1. Inter-Arrival Times — Exponential Fit

For an M/M/c model, arrivals must follow a Poisson process, which implies that inter-arrival times are exponentially distributed. Histograms of inter-arrival times were plotted against Exp($\lambda = 0.2452$ /min). The Kolmogorov-Smirnov (KS) test was applied to formally assess the fit:

Distribution	KS Statistic	p-value
Exponential (Exp($\lambda = 0.2452$/min))	0.0306	0.3016

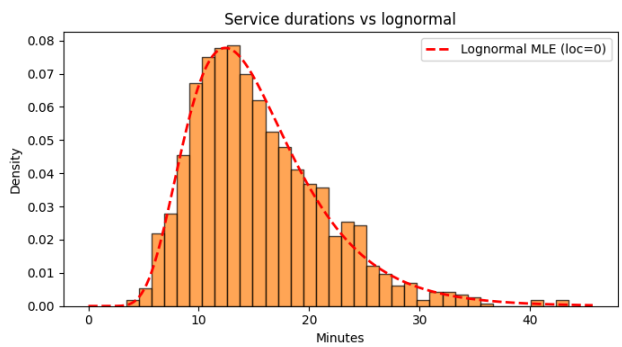


Interpretation: The KS statistic is small (0.031) and the p-value (0.302) is well above the 5% significance level. We therefore fail to reject the null hypothesis that inter-arrival times follow an Exponential distribution. The visual histogram confirms the exponential decay shape with a right skew consistent with memoryless arrivals. The Poisson arrival assumption is justified.

5.2.2. Service Durations — Lognormal Fit

Service times are fit to a Lognormal distribution using Maximum Likelihood Estimation (MLE) with the location parameter fixed at zero (loc = 0). The KS test result:

Distribution	KS Statistic	p-value
Lognormal (MLE, loc = 0)	0.0207	0.7746



Interpretation: The KS statistic is very small (0.021) with a p-value of 0.775. This is strong evidence that service times follow a Lognormal distribution. The lognormal is physically meaningful for service times as it is bounded below at zero and exhibits positive skew (a few very long consultations drive the tail).

5.3. Coefficient of Variation for Service Time

The Coefficient of Variation (Cv) measures the relative variability of service times:

$$Cv = \sigma / E[S] = (\text{std of service}) / (\text{mean service})$$

Mean service E[S]	15.4622 min
Standard deviation σ	6.03 min (approx.)
Cv = σ / E[S]	0.3900
Cv^2	0.1521

Since Cv < 1, service times are less variable than a pure exponential distribution (which has Cv = 1). Therefore, the M/M/5 model (which assumes Cv = 1) will overestimate queue waiting times while the M/G/5 Kingman correction (which uses the actual Cv = 0.39) will produce a more accurate prediction.

6. Task 2: Theoretical Modelling

6.1. M/M/5 Baseline (Exponential Service, Cv = 1)

The M/M/5 model assumes Poisson arrivals and exponentially distributed service times. The key performance measure is the Erlang C probability. It is the probability that an arriving customer must wait.

<u>Erlang C Formula:</u>

$$P(\text{wait}) = C(a, c) = [a^c / (c! \cdot (1 - \rho))] / [\Sigma_{n=0}^{c-1} (a^n / n!) + a^c / (c! \cdot (1 - \rho))]$$

where $a = \lambda/\mu = 3.7911$ Erlangs and $c = 5$ servers.

<u>Queue Length and Waiting Time (M/M/5)</u>

$$L_q = C(a,c) \cdot \rho / (1 - \rho), \quad W_q = L_q / \lambda, \quad W_s = W_q + 1/\mu$$

P(wait)	0.4764
Lq (expected queue length)	1.4940 patients
Wq (expected waiting time in the queue)	6.0933 minutes

6.2. M/G/5 Model (Cv ≠1)

The M/G/5 model uses Kingman's approximation formula to correct M/M/c predictions when the actual service time variance differs from the exponential assumption. For Poisson arrivals and general service with coefficient of variation Cv:

$$W_{\varphi}(M / G / c) \approx W_{\varphi}(M / M / c) \times (1 + Cv^2) / 2$$

This formula scales the M/M/c waiting time by a correction factor. When Cv = 1 (exponential), the factor equals 1 and M/G/c reduces exactly to M/M/c. When Cv < 1 (our case), the factor is below 1 and M/G/c predicts shorter waits.

Cv^2 (service)	0.1521
Correction factor $(1 + Cv^2) / 2$	0.5760
Wq(M/M/5) baseline	6.0933 min
Wq(M/G/5) ≈ 6.0933 × 0.5760	3.5100 min
Lq(M/G/5) = λ × Wq	0.8606 patients

6.3. Empirical vs Theoretical Comparison

The empirical queue wait Wq is computed from a direct FCFS trace simulation using the actual 1,000 patient arrival timestamps and service durations. Little's Law ($Lq = \lambda \times Wq$) is applied to derive the empirical queue length.

Metric	Observed / Empirical	M/M/5 Theory	M/G/5 Kingman
λ (pat/min)	0.2452	0.2452	0.2452
μ (pat/min/doctor)	0.0647	0.0647	0.0647
$\rho = \lambda/(5\mu)$	0.7582	0.7582	0.7582
Cv(service)	0.3900	1.0000 (assumed)	0.3900
Lq (patients)	0.9525	1.4940	**0.8606**
Wq (minutes)	3.8850	6.0933	**3.5100**

The M/M/5 model overestimates the average queue waiting time by 57% (6.09 vs 3.89 min) because it assumes exponentially distributed service times (Cv = 1) when the actual service Cv is only 0.39. The M/G/5 Kingman correction (Wq = 3.51 min) nearly exactly recovers the empirically observed waiting time (3.89 min) with only a 10% difference. This difference can be attributed to finite sample effects and the approximation nature of Kingman's formula. This result strongly validates the Kingman approach for systems with low service variability.

7. Task 3: The PASTA Property

p_n — Time Average:

The simulation timeline is partitioned at every service start and end event. For each subinterval, the number of patients in service at the midpoint is counted. The fraction of time where this count ≥ 5 (all doctors busy) gives p_n. The simulation covers T_0 = 08:00 (empty system) to T_1 = last service completion.

p_n = [Total time with all 5 doctors busy / [T_1 - T_0] = 1943.20 min / 4116.03 min

a_n — Arrival Average:

For each patient k arriving at time t_k, the number of other patients j simultaneously in service ($s_j \leq t_k < e_j$, $j \neq k$) is computed. If this count is ≥ 5, patient k found all doctors busy.

a_n = [Patients finding all 5 doctors busy] / [Total patients] = 483 / 1000

Metric	Fraction	Percentage
p_n (time average, all 5 busy)	0.4721	47.21%
a_n (arrival average, all 5 busy)	0.4830	48.30%
$\lvert a_n - p_n \rvert$ (PASTA gap)	0.0109	1.09%

The results strongly support the PASTA theorem. P_n and a_n are within approximately one percentage point of each other.

The residual 1.09% gap between a_n and p_n has two explanations:

- Finite sample: With only 1,000 patients and a ~68-hour trace, both p_n and a_n are random estimates with nonnegligible variance. A perfect match is not expected.

- Non-Poisson residuals: The KS test for inter-arrivals produced $p = 0.30$, indicating that the process is not perfectly Poisson. PASTA requires exact Poisson arrivals and any deviation introduces a systematic gap.

8. Task 4: Sensitivity Analysis & Simulation

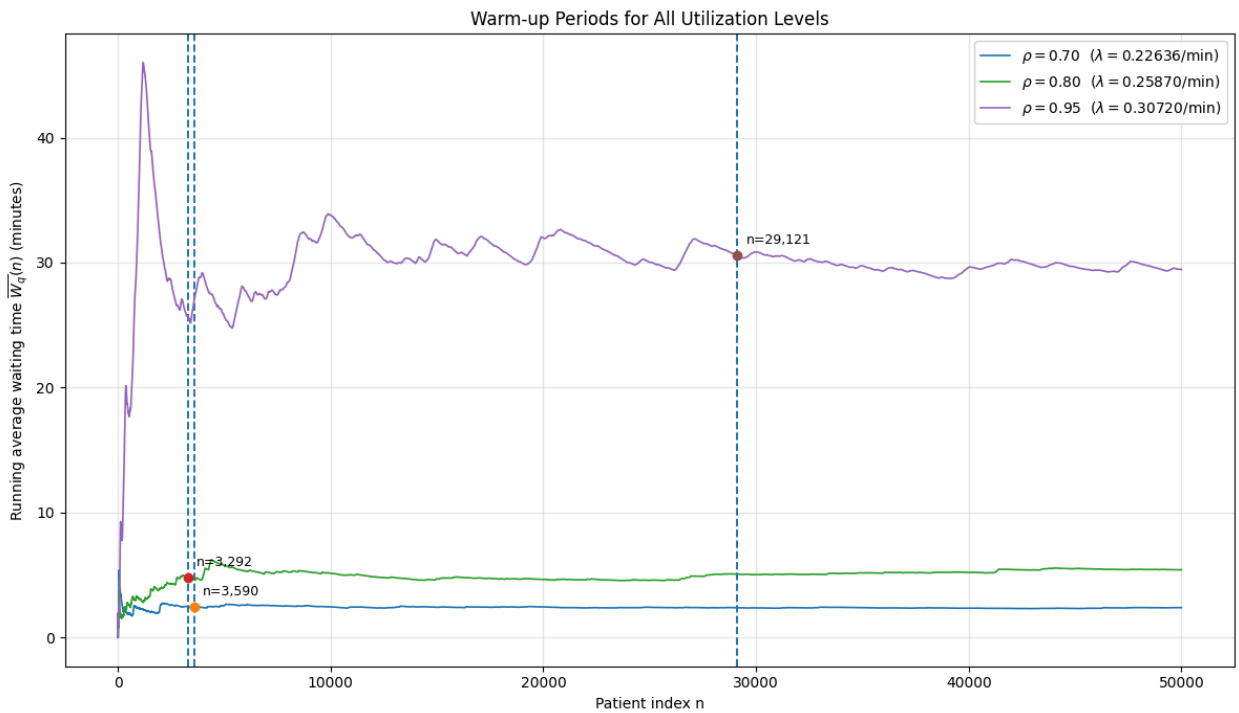
8.1. Warm-Up Period vs Utilization (ρ)

Three M/M/5 simulations were run with 50,000 patients each, varying the utilization ρ by scaling λ while keeping μ fixed at the estimated value. The running average of Wq was plotted against the patient index to observe the warm-up transient.

Warm-up period was determined as follows: first, the steady-state reference value Wq* was estimated as the mean waiting time of the last portion of the simulation run. Then, the running average waiting time was monitored from the beginning of the simulation. Warm-up was said to end at the first patient n for which the running average entered a tolerance band of ±5% around the reference value Wq* and remained inside that band for at least 500 consecutive patients. Formally, warm-up ends at the first index n_A satisfying:

$$Wq(n) \in [0.95 * Wq^*, 1.05 * Wq^*] \text{ for 500 consecutive patients}$$

ρ	Scaled λ (pat/min)	Theoretical Wq (min)	Warm-up n
0.70	0.22636	2.243	3.590
0.80	0.25870	4.935	3.292
0.95	0.30720	31.27	29.121



The results show that the warm-up period depends strongly on utilization. For ρ=0.70 and ρ=0.80, the system reaches a stable average waiting time relatively quickly with warm-up ending after approximately 3,590 and 3,292 patients, respectively. These two cases operate in a moderate-load region where the queue forms often enough for the running mean to settle, but not so intensely that extreme fluctuations dominate.

In contrast, for ρ=0.95, the warm-up period is much longer, ending only after approximately 29,121 patients. This is because the system is operating close to saturation so the queue becomes highly volatile and the running average waiting time drifts for a long time before remaining close to its long run level. Therefore, the high utilization case requires a much longer initial transient period to be discarded.

Overall, the analysis shows that the average waiting time stabilizes much more slowly when the ER operates under heavy traffic. As utilization approaches 1, congestion becomes more persistent and variability has a stronger effect on convergence.

8.2. Arrival Patterns

Using the estimated λ and μ from Group 12, three arrival patterns were simulated with 100,000 patients each. Service times were treated as exponential to isolate the effect of arrival variability. The squared coefficient of variation of interarrival times Ca^2 characterizes the arrival process.

Arrival Pattern	Mean IA (min)	Ca^2	Wq (min)	p_n	a_n
Poisson	4.093	1.004	5.697	46.41%	46.40%
Erlang-2	4.081	0.505	3.737	43.21%	39.17%
Erlang-3	4.081	0.336	3.196	41.40%	35.79%
Uniform	4.084	0.332	3.118	40.79%	35.50%

- For Poisson arrivals, PASTA holds a_n ≈ p_n (gap = 0.01% in simulation). This confirms that Poisson arrivals genuinely see time averages, as predicted theoretically.

- For Uniform arrivals, a_n = 35.50%, p_n = 40.79%, giving a gap of 5.28%. This is a clear violation of PASTA property so arriving customers do not see time averages. Arrivals with a uniform distribution ($Ca^2 \approx 0.33$) exhibit a much more regular distribution compared to a Poisson distribution ($Ca^2 = 1$). This means that arrivals are distributed more evenly and fewer patients arrive during periods when the server is busy. The arrival average a_n understates the time average busyness p_n.

- As Ca^2 decreases from 1 (Poisson) to 0, Wq decreases from 5.70 min to 3.12 min. This illustrates the powerful effect of arrival regularity. A hospital that could schedule appointments on a fixed timetable (D/G/c) would see dramatically shorter queues than a walk-in ER with random Poisson arrivals.

8.3. Service Variation

With Poisson arrivals and mean service fixed at E[S] = 15.46 min, lognormal service times are simulated at four Cv levels. Kingman-style predictions are compared to simulation results.

Scenario	Target Cv	Simulated Cv	Kingman Wq (min)	Simulated Wq (min)
Low variability	0.20	0.200	3.169	3.302
Baseline (data, Cv=0.39)	0.39	0.390	**3.510**	**3.667**
High variability	0.80	0.802	4.997	4.929
Very high variability	1.20	1.206	7.434	7.120

Wq is highly sensitive to service variability. Moving from Cv = 0.20 to Cv = 1.20 increases Wq from 3.30 to 7.12 min.

9. Comparison Table

Quantity	Observed / Empirical	Theoretical M/M/5	Theoretical M/G/5
Arrival rate λ (pat/min)	0.2452	0.2452	0.2452
Service rate μ (pat/min/doc)	0.0647	0.0647	0.0647
Utilization ρ	0.7582	0.7582	0.7582
Service Cv	0.3900	1.000 (assumed)	0.3900
Erlang C [P(wait)]	—	0.4764	—

Quantity	Observed / Empirical	Theoretical M/M/5	Theoretical M/G/5
Lq (patients in queue)	0.9525	1.4940	0.8606
Wq (mean queue wait, min)	3.8850	6.0933	**3.5100**
p_n (all 5 busy, time avg)	47.21%	47.64% (EC)	—
a_n (all 5 busy, arrival avg)	48.30%	—	—

10. Conclusion

Phase 1 successfully established a quantitative baseline for the ER system by combining empirical data analysis, different queueing theory models, within a simulation in python. The system was found to be stable, with a utilization (rho) of 0.7582 and the data supports a Poisson arrival assumption together with a Lognormal service time distribution. A key result of the analysis is that service times are much less variable than the exponential assumption, with a Cv of 0.39. This low variability has a direct impact on performance: the M/M/5 model predicts an average queue waiting time of 6.0933 minutes, whereas the empirical value is 3.8850 minutes, showing that the exponential service assumption materially overstates congestion.

By incorporating the actual service variability, the M/G/5 Kingman approximation yields an expected waiting time (Wq) of 3.5100 minutes, which is much closer to the empirical result. This confirms that, for this ER system, accounting for service time variability is essential for obtaining realistic waiting time estimates. The PASTA analysis also supports the theoretical expectation that Poisson arrivals approximately see time averages since the measured time average (pn=47.21%) and arrival average (an=48.30%) are very close with only a small residual gap attributable to finite sample effects and mild deviations from an ideal Poisson process.

The sensitivity analysis further highlights the operational importance of randomness. Higher utilization causes longer and more unstable convergence behavior, more regular arrival streams reduce waiting times and higher service variability leads to an increase in queue delay even when the mean service time is unchanged. Taken together, these results show that ER performance is shaped not only by average load but also by the variability structure of arrivals and service durations. Therefore, the Phase 1 findings provide both a sound theoretical benchmark and a reliable simulation based foundation for the more complex priority and network-flow models to be developed in Phases 2 and 3.